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// Generics_Day_01.png
// This code is for a generic function that takes a list of integers and returns a list of squares.
// It is a simple example of how to use generics in Java.

// Importing the ArrayList class from the java.util package
import java.util.ArrayList;

// Defining a generic function that takes a list of integers and returns a list of squares
public static ArrayList<Integer> squareList(ArrayList<Integer> list) {
    // Creating a new ArrayList to store the squares
    ArrayList<Integer> squares = new ArrayList<>();

    // Iterating over the input list and calculating the squares
    for (Integer num : list) {
        // Calculating the square of the number
        Integer square = num * num;

        // Adding the square to the squares list
        squares.add(square);
    }

    // Returning the list of squares
    return squares;
}

// Testing the function with a sample list of integers
public static void main(String[] args) {
    // Creating a sample list of integers
    ArrayList<Integer> numbers = new ArrayList<>();
    numbers.add(1);
    numbers.add(2);
    numbers.add(3);
    numbers.add(4);
    numbers.add(5);

    // Calling the squareList function with the sample list
    ArrayList<Integer> squares = squareList(numbers);

    // Printing the resulting list of squares
    System.out.println("Squares: " + squares);
}

// Output: Squares: [1, 4, 9, 16, 25]
```